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Iberia in the Cretaceous: new paleomagnetic results from Portugal

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Abstract

From the Lusitanian-basin and from Algarve, Toarcian through Cenomanian sediments (16 localities, 170 samples) and upper Cretaceous igneous rocks (4 localities, 31 samples) were subjected to paleomagnetic analysis. Sedimentary localities yielded well-defined characteristic magnetizations, with north-westerly declinations, both before and after tilt correction. Igneous rocks exhibited both north-westerly and north-easterly declinations.

The paleomagnetic directions of the sediments, when evaluated together with earlier published paleomagnetic results from the Jurassic–Cretaceous sediments of the Lusitanian basin and Algarve, suggest that Iberia rotated about 26° counterclockwise, with respect to stable Europe, after Cenomanian, and before the emplacement of the Lisbon volcanics (around 70 Ma). In the Cretaceous paleomagnetic data set, increased by the results of the present study, there is no evidence for earlier and additional counterclockwise rotation during Cretaceous.

While the net post-Permian and post-Cenomanian counterclockwise rotations of Iberia with respect to present north are practically of the same angle, they are different with respect to stable Europe. The difference implies clockwise rotation of stable Europe with respect to Iberia, before Cretaceous.

The results of this study from igneous rocks are relevant to the late Cretaceous–early Tertiary movements of Iberia. We suggest that the two igneous bodies with north-westerly declinations (one belonging to the Lisbon volcanics, the other a "Salema intrusion") were intruded before the post-Cenomanian counterclockwise rotation of Iberia;

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the north-easterly declinations exhibited by two syenite sites and three cross-cutting dykes from the Monchique intrusion are indicative of a small clockwise rotation, probably connected to the Pyrenean mountain building process.
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1. Introduction

Since Carey's suggestion (1958) that the Bay of Biscay opened in the Eocene, due to CCW rotation of Iberia relative to Europe, there have been a number of paleomagnetic studies about the "largest Tethyan block" (Van der Voo, 1993). Van der Voo (1967) published the first solid evidence for the rotation of Iberia, 35° CCW since the Triassic, which was interpreted as due to the opening of the Bay of Biscay (Van der Voo, 1969).

Subsequent studies from stable Iberia suggested that the position of Iberia was not constant during the Cretaceous. Results from igneous rocks seemed to suggest complicated rotations, both in the CCW and CW sense, before Iberia became eventually attached to Europe (Storetvedt et al., 1987, 1990). Paleomagnetic data from Jurassic–Cretaceous sediments (Galdeano et al., 1989; Moreau et al., 1997) indicated only CCW rotations during the Cretaceous. Galdeano et al. (1989) described a 27° CCW rotation of Iberia in the 125–110 Ma interval, and a second one (14° CCW) between 110 and 83 Ma. Moreau et al. (1997) postulated a CCW rotation of 22° ± 14°, between 132 and 124 Ma.

Estimations of the time, sense and magnitude of the rotations involved in the opening process are also known from the analysis of the magnetic anomalies in the Bay of Biscay. Scotese et al. (1988) distributed a total of 35° CCW rotation between 92–84 Ma (30°) and 84–66 Ma (5°); Srivastava et al. (1990) suggested a single-phase CCW rotation of 30° in the 112–83 Ma interval.

The above studies demonstrate that the (late Jurassic) Cretaceous geodynamic history of the Iberian block is far from being well known. In order to improve the paleomagnetic control, we decided to obtain new paleomagnetic data from the Mesozoic sediments of two basins, which are in autochthonous position on the Hesperian massif, the Lusitanian basin and Algarve. We also studied some igneous sites from both basins, so that possible overprinting of sediments during igneous activity could be detected.

2. Geological framework

During the latest Hercinian phases, the Hesperian Massif was subjected to intense fracturing. Mesozoic sedimentation took place on the western (Lusitanian basin) and southern (Algarve) margin of the Massif, in depressions related to the opening of the Atlantic (Wilson, 1975; Boillot et al., 1978; Ribeiro et al., 1979). The Mesozoic fill of the Lusitanian basin is represented by five disconformity-bounded sequences (Pena dos Reis et al., 1992): Late Triassic–Callovian, Oxfordian–Berriasian, Valanginian–Early Aptian, Late Aptian–Early Campanian and Late Campanian–Maastrichtian. In Algarve, there are four megasequences (Manuppella, 1988; Martins, 1991) in the Triassic–Jurassic, and three in the Cretaceous. The Cretaceous disconformities are of Valanginian and Early Aptian age, respectively. In both basins, fluviatile–deltaic, and open shelf to hemipelagic deposits occur in different proportions in different cycles, and are represented by clastic material of different grain sizes, limestones and dolomites.

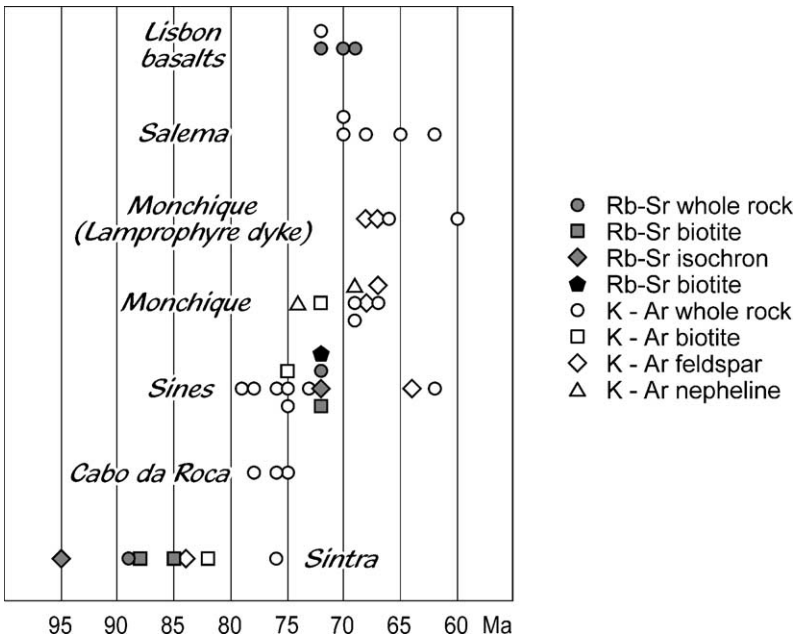


Fig. 1. Summary of the available isotope ages for the late Cretaceous igneous rocks from Portugal. Data with larger than 10 Ma analytical error are omitted. References are found in the text.

During the late Cretaceous, Sintra (including Cabo da Roca), Sines and Monchique massifs were emplaced, along a deep-seated NNW–SSE trending fault. Direct stratigraphic evidence puts the age of the Sintra pluton between late Cenomanian and Oligocene, while the stratigraphic gap is much wider for the Sines (Jurassic to Pliocene) and Monchique (post-Carboniferous) complexes. In addition, important igneous activity is known during Turonian–Eocene (Lisbon volcanics) in the Lusitanian basin, and there are intrusions of moderate sizes in Algarve, the "Salema intrusions" (petrological affinity connects them to the Monchique complex) which cut Triassic and Jurassic sediments. Radiometric data for the massifs are also available, which are Rb–Sr whole rock isochron, Rb–Sr model, and K/Ar ages (Bonhomme et al., 1961; Mendes, 1968; Canilho, 1971; Rock, 1976; Abranches and Canilho, 1981; Canilho and Abranches, 1982; MacIntyre and Berger, 1982; Storetvedt et al., 1987; Storetvedt et al., 1990). For the Lisbon basalts the last two (Ferreira and Macedo, 1979; Alves et al., 1980), for the Salema intrusions, only the last (Storetvedt et al., 1990) method was used. The data, with less than 10 Ma analytical error, fall into the 80–60 Ma interval, except Sintra, which seems to be somewhat older (Fig. 1). Despite of the fairly large number of radiometric data, the succession of the igneous rocks is poorly constrained, since the methods used in obtaining the data were very varied (Fig. 1).

3. Paleomagnetic sampling and laboratory procedure

From the Lusitanian basin and from Algarve, we collected well-stratified sediments of Toarcian through Cenomanian age at 16 localities as far away from the igneous massifs as possible. However, one locality (locality 11, Fig. 2) in the Lusitanian basin was close to a narrow dyke of the Lisbon volcanics (locality

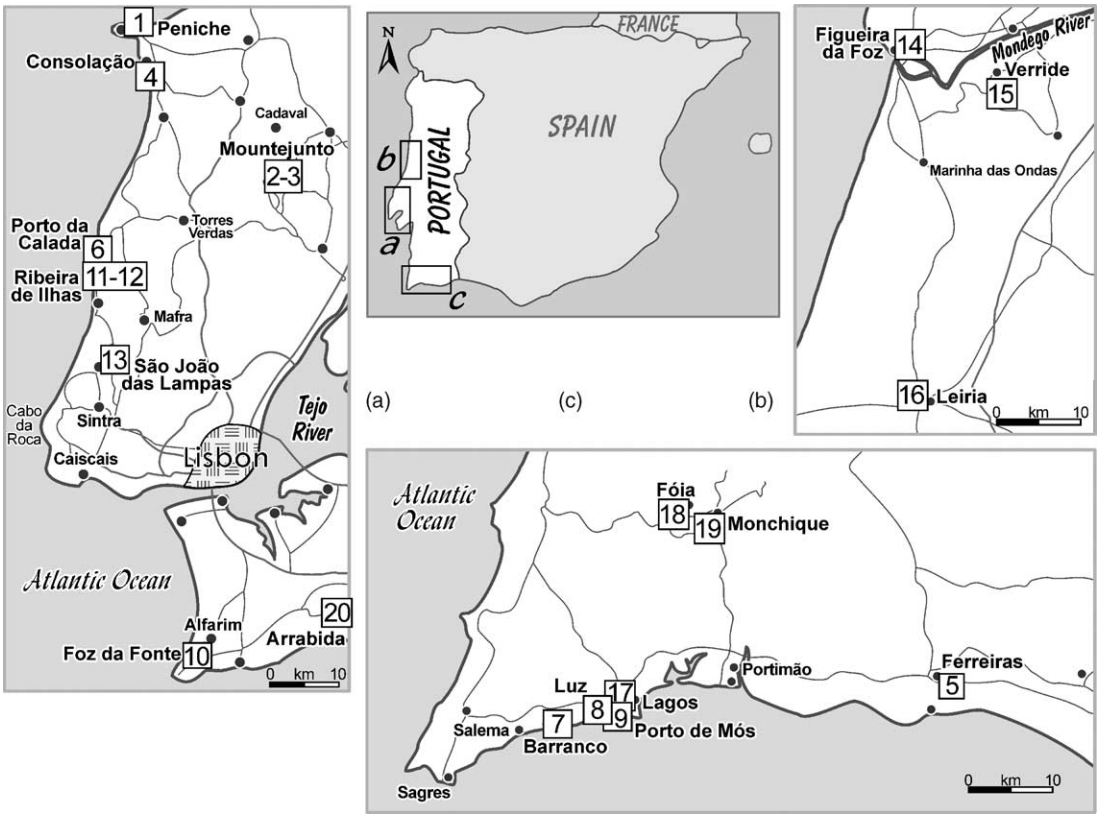


Fig. 2. Paleomagnetic sampling localities in the Lusitanian Basin (a and b) and in Algarve (c). Insert shows the sampling areas on the contour map of Iberia.

12, Fig. 2) and in Algarve, localities 8 and 9 (Fig. 2) were not far from a small "Salema intrusion" (locality 17, Fig. 2). In such cases, both the sediments and the igneous rocks were sampled. We also drilled several samples from the Monchique syenite (two sites) and from three cutting dykes, in order to have comparison with the "Salema intrusion" of our study (locality 17), since Storetvedt et al. (1990) did not publish paleomagnetic directions for localities or sites, but only defined components for the whole Monchique igneous complex (including Salema intrusions).

The samples were drilled in the field and oriented magnetically. Standard-size specimens were cut from the cores in the Paleomagnetic Laboratory of Eötvös Loránd Geophysical Institute. NRM and susceptibility of each specimen was measured in the same laboratory, using JR-4 and JR-5A spinner magnetometers and KLY-2 kappabridge, respectively. At least one specimen per sample was demagnetized in increments, either thermally (in this case susceptibility was monitored) or with alternating field (AF) method. Orthogonal projections of the demagnetization curves were analyzed for linear segments (method by Kirschvink, 1980), thus defining the NRM component(s). Site or locality mean paleomagnetic directions were calculated (method by Fisher, 1953), in the geographic and for the sediments also in the tectonic system. The localities and sites with statistically well-defined mean directions are shown in Table 1. Magnetic mineralogy experiments included susceptibility versus temperature (continuous heating) and

Table 1
New paleomagnetic results from the Jurassic-Cretaceous of the Lusitanian Basin (localities marked with L) and from Algarve (localities marked with A)

Locality		*Age* Lithology	n/no	D (°)	I (°)	k	α_{95} (°)	D_C (°)	I_C (°)	k	α_{95} (°)	Dip
1L	Peniche	*Toarcian*	7/8[a]	21	+60	75	7	4	+84			210/24
	PH 87–94	Marl	8/8	330	+51	31	10	296	+59			
2L	Montejunto	*Oxfordian*	14/14	343	+38	159	3	330	+39	77	7	variable
	PH 61–74	Limestone										
3L	Casal da Remeda	*Oxfordian*	11/12	345	+47	50	7	270	+58			214/50
	PH 75-86	Marl										
4L	Consolacão	*Kimeridgian*	8/9[a]	1	+59	92	6	18	+61			120/10
	PH 95-103	Marl	9/9	330	+43	51	7	337	+51			
5A	Ferreiras	*Kimeridgian*	7/8	339	+43	165	5	335	+39			297/6
	PH 193–200	Limestone										
6L	Porto da Calada	*Berriasian*										
	PH 24–37[b]	Marl	13/14[b]	327	+33	18	10	327	+30			8/3
	PH 24–31		9/9	338	+38	193	4	339	+36			
	PH 32–37		4/5	119	−20	27	18	118	−18			
7A	Barranco	*Berriasian*	6/11	355	+28	188	5	355	+48			180/20
	PH 152-162	Calcareous marl	5/11	326	+28	21	17	318	+45			
8A	Luz	*Aptian*										
	PH 168–172	Grey marl	5/5	324	+31	150	6	330	+39			108/12
	PH 173–177	Red marl	5/5	333	+35	224	5	341	+42			
9A	Porto de Mós	*Late Aptian*	4/9	336	+42	219	6	335	+46			172/4
	PH 184–192	Marl										
10L	Foz da Fonte	*Albian*	11/18[a]	1	+44	40	7	357	+44			10/2
	PH 1–17	Marl	9/18	339	+49	65	6	340	+48			
11L	Ribeira de Ilhas	*Albian*	7/10	325	+50	34	10	316	+56			194/9
	PH 38–47	Marly limestone										
12L	Ribeira de Ilhas	Basalt dyke	5/7	343	+40	367	4					
	PH 48–54											
13L	São João das	*Cenomanian*	6/6[a]	350	+54	1607	2	343	+51			290/7
	Lampas	Limestone	3/6	327	+35	28	24	324	+29			
	PH 55–60											
17A	Lagos	Intrusion	6/6	150	−40	219	4					
	PH 178–183											
18A	Fóia											
	PH 144–148	Dyke	5/5	180	−51	44	12					
	PH 149–151	Syenite	3/3	215	−48	62	16					
19A	Monchique											
	PH 134–136	Dyke	2/3	39	+36	–	–					
	PH 137–140	Dyke	4/4	220	−70	409	6					
	PH 141–143	Syenite	3/3	197	−54	306	7					

Localities 1–11 and 13 are sediments, 12 and 17–19 are igneous rocks. Key: n/no: number used/collected samples (the samples are independently oriented cores); D, I (D_C, I_C): declination, inclination before (after) tilt correction; k and α_{95}: statistical parameters.

[a] Overprint.

[b] Locality mean paleomagnetic direction calculated from samples both with normal and reversed polarities.

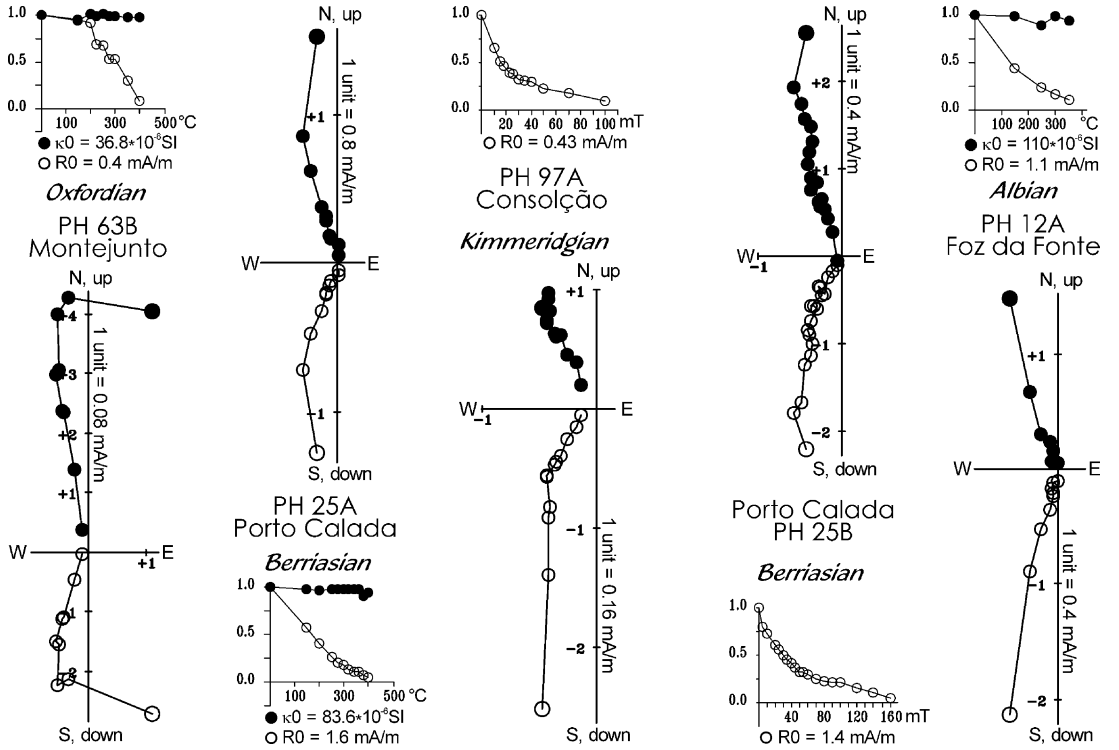


Fig. 3. Typical demagnetization curves for sediments from the Lusitanian basin. Original Zijderveld diagrams and intensity/susceptibility vs. temperature or intensity vs. demagnetizing field curves. In the Zijderveld diagrams, dots/open circles: projection of the NRM in the horizontal/vertical plane; in the others susceptibility: dots, NRM intensity: circles.

IRM acquisition experiments, and the stepwise thermal demagnetization of the three-component NRM (method by Lowrie, 1990).

4. Results

With the exception of localities 14–16 and 20 (Fig. 2) with extremely weak natural remanences, the sediments, which were grey (except one red bed at locality 8A) and devoid of any visible alteration, are characterized by very good demagnetization behavior on thermal treatment. AF demagnetization was also efficient in some cases (Figs. 3 and 4). There are specimens, in which two components were identified; in others, the NRM was practically single component. The characteristic remanences invariably have northwesterly declination with positive inclination (there is only one site, 6L, Table 1, with reversed polarity) in the geographic system (Figs. 3 and 4 and Table 1), showing moderate variation between localities of different ages or far apart. As a rule, NRM is practically demagnetized by 400 °C. Exception is the red marl (Fig. 4, PH 175A), where the NRM signal disappeared only at 600 °C.

Proportion of softer and harder components in the IRM acquisition curves (Fig. 5) is variable, but the soft magnetic component with unblocking temperature of 580 °C is always dominant, indicating that the main magnetic mineral is magnetite. The decay curves of medium and hard components run parallel to

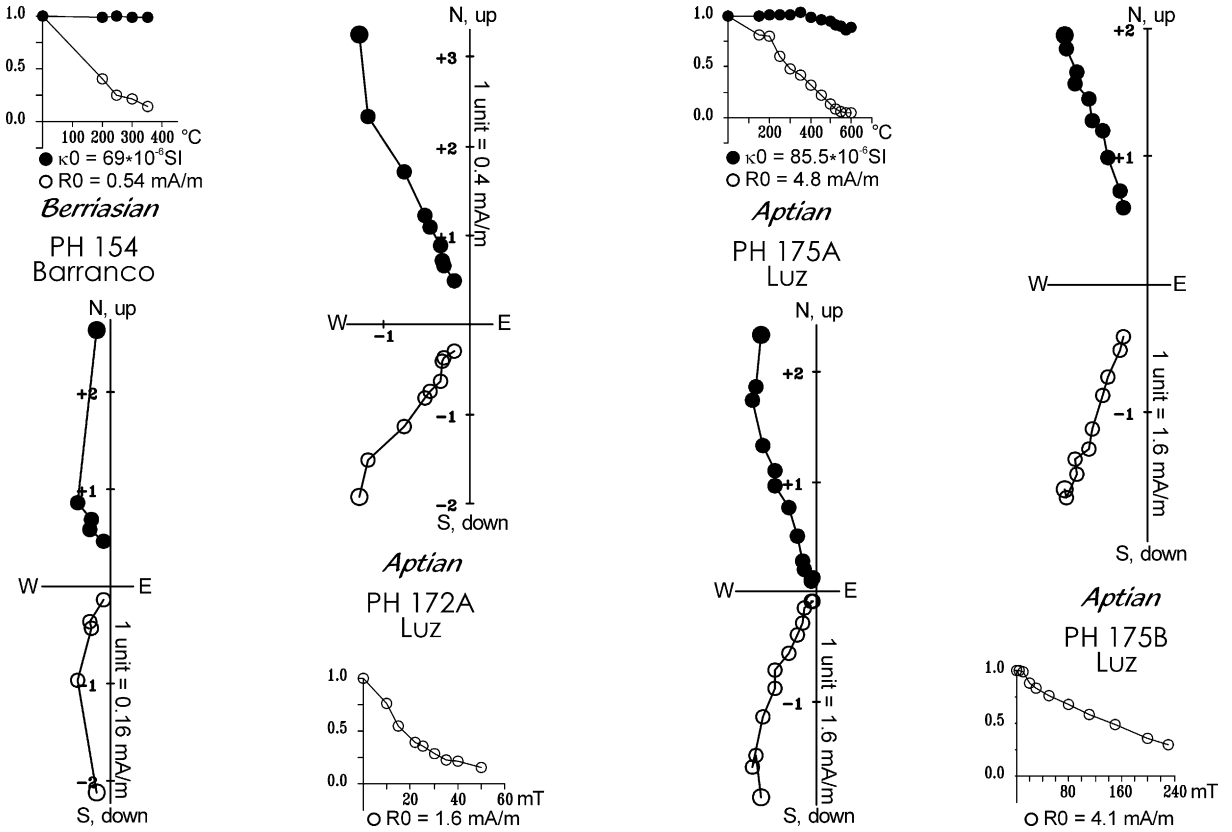


Fig. 4. Typical demagnetization curves for sediments from Algarve. Key as for Fig. 3.

that of the soft component, but a small part of the IRM intensity survives the Curie point of magnetite. Susceptibility increases in most samples of pre-Aptian age from 400 to 450 °C on (Fig. 5, PH 63A), due to formation of magnetite or maghemite from iron sulphide. Curie points of the magnetic minerals (method of continuous measuring of susceptibility on heating) were generally not measurable, due to conversion of iron sulphide to oxide on continuous heating. In specimens without iron sulphide, magnetite was identified (e.g., PH 25).

Igneous specimens revealed similar components on both AF and thermal demagnetization (Fig. 6). The results of demagnetization and component analysis are shown in Table 1 (12L, 17A, 18A, 19A). It is worth mentioning that the “Salema dyke ” in Algarve has reversed polarity NRM (unlike the sediments nearby) and indicate CCW rotation (like the sediments nearby), but the syenite as well as the dykes in the Monchique intrusion are of different directions.

5. Discussion and conclusions

Stratigraphic ages for the studied sediments vary between Toarcian and Cenomanian, but the NRMs represent a narrower time interval. The reason is that some of the studied strata seem to have acquired

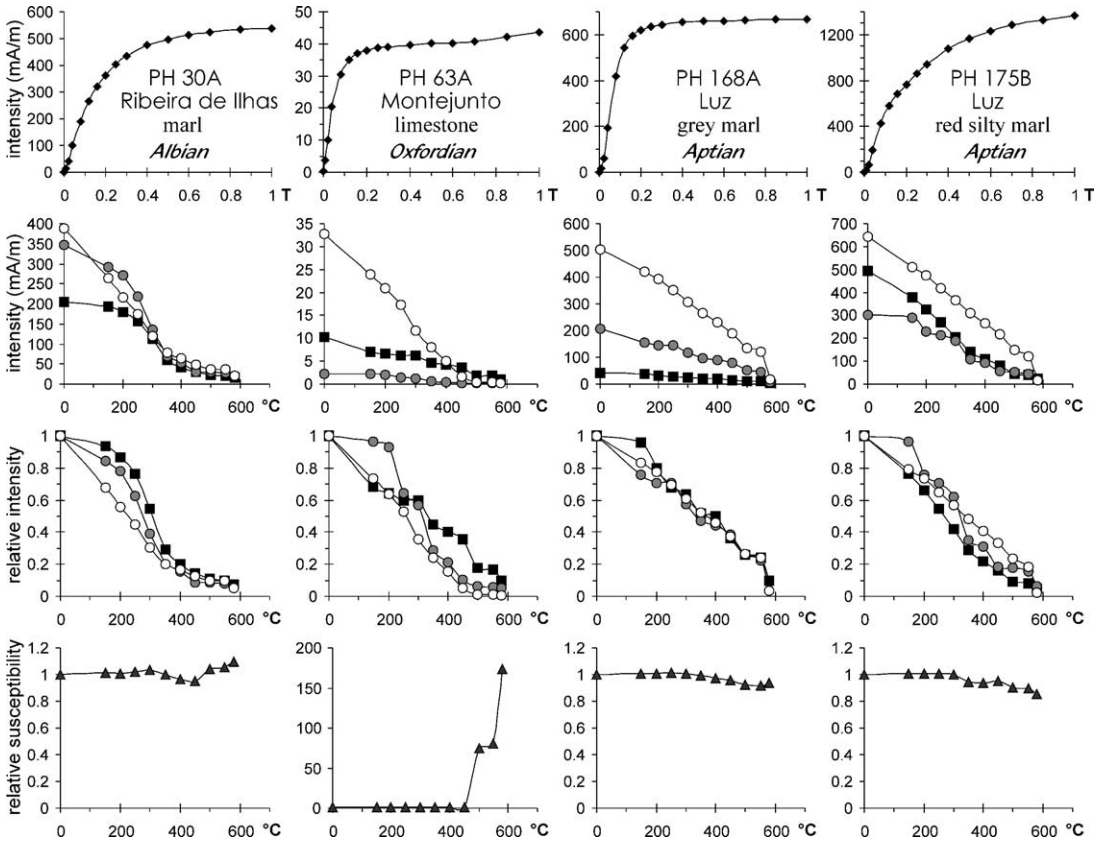


Fig. 5. Magnetic mineralogy as shown by IRM acquisition behavior and the thermal demagnetization of the three-component IRM. IRM acquisition curves (upper row), the behavior of the three-component IRM (Lowrie, 1990) on thermal demagnetization and the same normalized to the initial value (second and third row, respectively) and the change in susceptibility on heating (lowermost row). The hard (squares), the medium hard (dots) and soft (circles) components of the composite IRM were acquired in fields of 1.0, 0.36 and 0.2 T, respectively.

remanence well after deposition. When we compare the locality mean directions before tilt corrections and after restoring the strata to the horizontal position, we can occasionally observe a dramatic shift in the directions (e.g., locality 3L, Table 1) but more often only a little change (e.g., locality 9A, Table 1), depending on the tilt angle. During untilting, the grouping of the paleomagnetic directions within locality remains unchanged, since the bedding attitudes at the respective localities are uniform. Exception is locality 2L (Table 1), with variable within- locality tilts. In this case we were able to constrain the age of the remanence with respect to the age of folding, with the conclusion that the NRM is of postfolding age. For the rest, classical fold test could not be carried out. According to the general practice, tilt test on regional level was used to constrain the age of the remanence. This test needs data from a fairly narrow time interval and from the same tectonic unit.

Since both the Lusitanian basin and Algarve represent stable Iberia, data from the two basins can be treated as from the same tectonic unit. The question is what are the properly selected age groups. The tilt test requires minimum three locality mean directions, but it is surely more robust when based on a larger

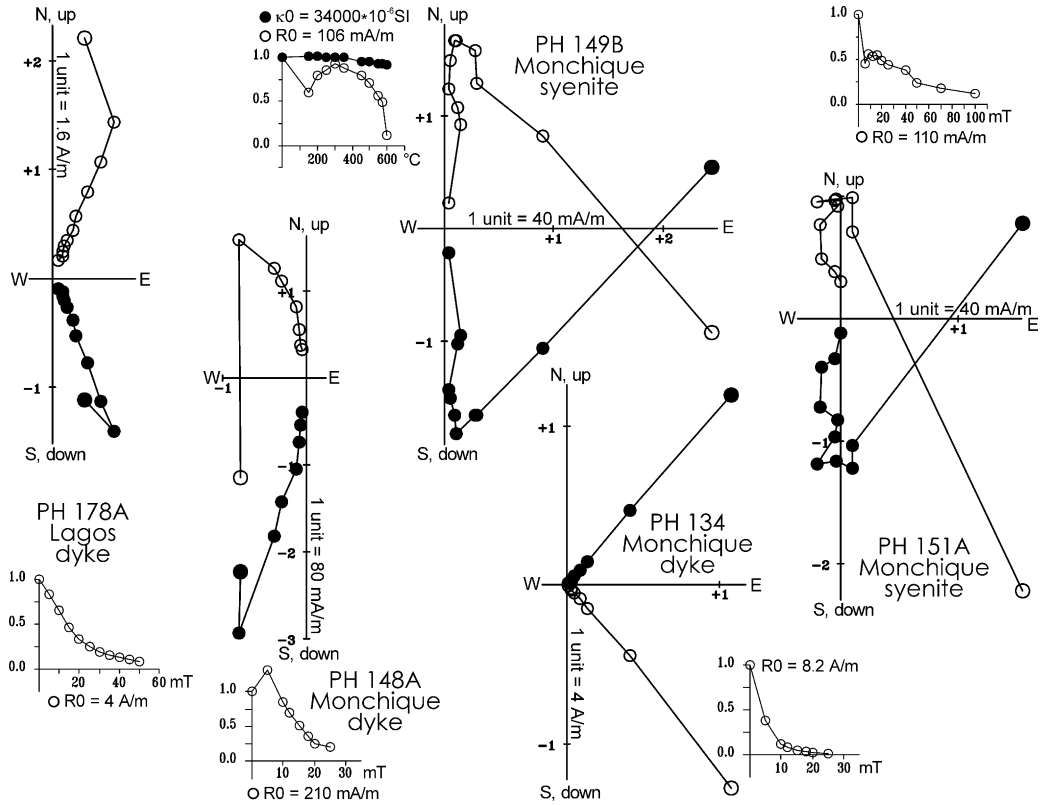


Fig. 6. Typical demagnetization curves of Cretaceous igneous rocks. Key as for Fig. 3.

set of data. That is why data shown in Table 1 were used in combination with previously published ones (Galdeano et al., 1989; Moreau et al., 1997). Moreover, to have enough data in one group, the overall-mean directions of Table 2 were not always combined by stages, which would be ideal, but sometimes for a pair of stages. In the latter case we considered the disconformities as boundaries between groups.

As Table 2 and Fig. 7 show, the overall-mean paleomagnetic directions calculated for the shortest possible time intervals (at their best Fisher statistics) scatter around a mean direction of $D = 334^\circ$, $I = 47^\circ$, without any definite trend during late Jurassic–early Cretaceous. A closer inspection reveals, that the paleomagnetic directions shown in Fig. 7 are not of the same quality. For the Aptian and for the Albian–Cenomanian groups, respectively, the tilt test is positive, i.e., the remanences are of pre-tilting age (Table 2). This implies first, that Iberia did not rotate between the Aptian and Cenomanian, second, that about 26° counterclockwise rotation must have occurred after the Cenomanian, relative to the present North.

The tectonic interpretation of the paleomagnetic directions obtained for the pre-Aptian sediments is less straightforward. Tilt test shows that the remanence is most likely of pre-folding (Oxfordian–Kimmeridgian group) or of synfolding (Portlandian, Berriasian, Valanginian–Hauterivian groups) age. Fortunately, the overall mean directions are practically the same in the present, in the tectonic coordinate systems and at the stage of unfolding with the best statistics. Therefore, the fundamental question is not the direction

Table 2
Summary of paleomagnetic results from sediments of the Lusitanian Basin and Algarve based on the present study, Galdeano et al. (1989), and Moreau et al. (1997)

Age	*N*	Before tilt correction				After tilt correction				Best Fisher statistics				
		D (°)	*I* (°)	*k*	α_{95} (°)	D_C (°)	I_C (°)	*k*	α_{95} (°)	% unfolding	*D* (°)	*I* (°)	*k*	α_{95} (°)
Oxfordian–Kimmeridgian	7	344	+39	75	7	320	+63	13	17	20	342	+44	114	6
Portlandian	3	334	+44	19	29	326	+55	25	25	60	330	+51	42	19
Berriasian	3	331	+35	40	18	324	+40	67	15	60	327	+39	85	13
Valanginian–Barremian	7	333	+35	14	17	326	+57	31	11	70	329	+50	48	9
Aptian	8	337	+44	19	13	340	+44	105	5	110	340	+44	113	5
Albian–Cenomanian	4	341	+45	28	18	336	+52	90	10	100	336	+51	90	10

Key as for Table 1.

to be used in tectonic interpretation, but the process and the time in which the original magnetic signal could have become partly or nearly completely overprinted.

Overprinting connected to late Cretaceous igneous activity can be ruled out for the following reasons. First, the localities selected by us were far from the large massifs, and so were those studied by Moreau et al. (1997). Second, the paleomagnetic directions for the massifs do not show any similarity to those

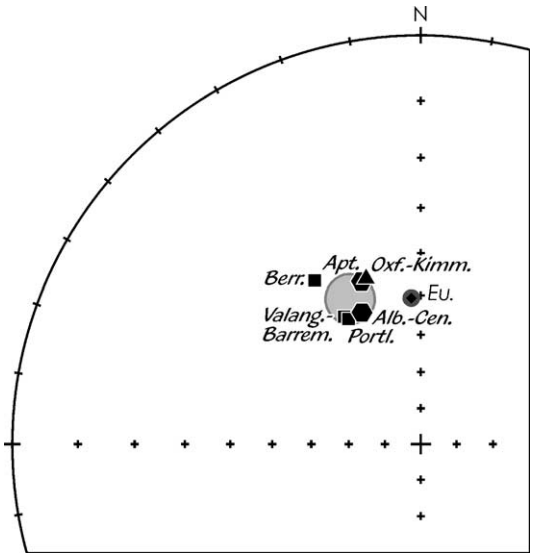


Fig. 7. Overall paleomagnetic mean directions for the late Jurassic–early Cretaceous sediments of stable Iberia. Stereographic projection, all vectors are lower hemisphere. Key: best statistics close to zero unfolding (triangle), close to 60–70% (square), close to 100% (hexagon), grey circle is the overall-mean Cretaceous paleomagnetic direction with α_{95}. *Eu* is the stable European reference direction for 110–80 Ma, with α_{95} (calculated from data compiled by Besse and Courtillot, 2002). Note that the main disconformities in both the Lusitanian basin and Algarve are of about 110 Ma.

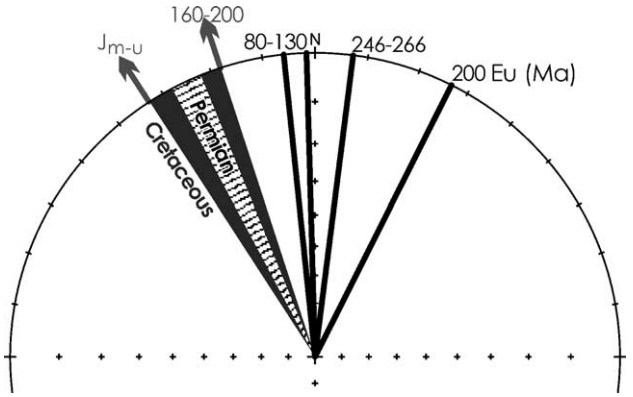


Fig. 8. Comparison of Permian (Van der Voo, 1993), Cretaceous (present study) and Jurassic (160–200 Ma, Schott et al., 1981, and J_{m-u}, Juárez et al., 1998) declinations shown by arrows and stable European reference declinations of corresponding ages (246–266 Ma, Van der Voo, 1993; 200 and 80–130 Ma, Besse and Courtillot, 2002, respectively). Note that Permian shaded declination range is contained within the wider Cretaceous range (black).

of the sediments used for evaluation. Remagnetization by small intrusions was a possibility ruled out, because of opposite polarities in the magmatic bodies and in the nearby sediments (e.g., localities 8A and 17A, Fig. 2).

As it was described earlier, the Mesozoic fills of both the Lusitanian basin and Algarve are subdivided into disconformity bounded megasequences. The most important of the disconformities is found close to boundary of the early and late Aptian. We suggest that remagnetization of the pre-Aptian rocks happened around this time, probably connected to a thermal event due to extensional tectonics. Such mechanism has been already invoked to explain remagnetization of Mesozoic carbonates in Iberia (e.g., Juárez et al., 1996, 1998; Dinarès-Turell and García-Senz, 2000). There is an other disconformity level, which could be candidate for the time of remagnetization, a post-Cenomanian one. However, this is highly unlikely, since the Aptian through Cenomanian groups preserved their primary remanences.

In summary, the joint evaluation of earlier published and present paleomagnetic data from the Jurassic and Cretaceous sediments of the Lusitanian basin and of Algarve suggests that Iberia rotated only once during the Cretaceous, before the emplacement of the large massives of Sintra and Monchique, and this happened in post-Cenomanian times. The overall-mean declination is 334°, indicating that the rotation was 26°, in the counterclockwise sense, with respect to the present north. Similar angles were measured for pre-Cretaceous, representing unquestionably stable Iberia (Fig. 8) and also for mid-late Jurassic sediments from the probably autochthonous part of the Iberian Range (Juárez et al., 1998).

Comparison of the declinations for Permian, Jurassic and Cretaceous rocks from stable Iberia with corresponding stable European reference directions reveals that the net post-Permian CCW rotation of Iberia relative to stable Europe is larger than the post-Cenomanian rotation. The difference is clearly due to CW rotation of stable Europe, before the Cretaceous (Fig. 8).

In contrast to the consistent paleomagnetic directions obtained from Jurassic–Cretaceous sediments, the available paleomagnetic directions for the Cretaceous igneous rocks from the Iberian autochthon are scattered. Their tectonic significance has been a matter of discussion (for a summary, read Van der Voo, 1993). The interpretation is handicapped also by the available radiometric ages since they do not permit to sort out exactly the succession of the late Cretaceous (Eocene) magmatic events. In our collection

of igneous rocks we did not really encounter complications, since the NRMs within each locality were consistent. By assuming that the Salema Intrusions are older than the measured K/Ar whole rock ages, the counterclockwise rotation indicated (and also the one exhibited by a dyke of the Lisbon volcanics, 12L in Table 1) could be connected to that of the Cenomanian and older sediments of this study. The small clockwise rotation observed on the Monchique igneous complex, however, calls for additional explanation.

To date, two scenarios are suggested by paleomagnetists for the end of Cretaceous. One envisages complex rotations, the second disregards "outliers" and suggests that the paleomagnetic directions from the late Cretaceous igneous rocks are practically aligned with the stable European reference direction (for more detail, see Storetvedt et al., 1990, Van der Voo, 1993). As the present study concentrated on sediments, we could not carry out a systematic paleomagnetic study on igneous complexes. Nevertheless, our results from the Monchique complex, which are indicative of a small clockwise rotation with respect to stable Europe seem to indicate that the late Cretaceous counterclockwise rotation of Iberia was followed by a small clockwise rotation (Lisbon volcanics, Van der Voo and Zijderveld, 1971, represent an intermediate step in this process). The approximately 26° clockwise rotation we observed on the Monchique igneous complex can represent the rotation of the whole Iberian block, causing compression and strike-slip movements in the Pyrenees.

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